

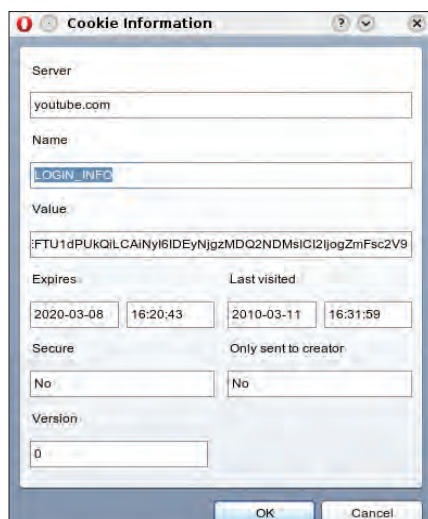
Is Big Brother Watching?

What about government security agencies around the world? Are they monitoring online activities too? Government agencies don't have enough resources to spare to monitor each person individually. The most common approach is to use software that can analyse a large amount of data, and automatically send an alert if any suspicious activity is detected.

attracting advertisers. However, analytics is relatively harmless, when compared to more intrusive techniques tracking.

Cookies are small files stored on the host computer to record the "state" of a session. This allows web sites to restore surfing activity to a previous point, for say a mail application, or a shopping cart on an e-commerce web site. Most web sites store a number of cookies on your machine, and some of these cookies can be used to track and identify individuals, giving rise to privacy concerns. Web sites are allowed to set "expiry dates" for the cookies that they save. Most cookies are valid for a few hours, a week or a month. A few cookies, however, last for a long time. This example shows a YouTube cookie that expires in 2020.

Most web sites also allow for third



A really long lasting YouTube cookie

party cookies to be saved on the host machine. This means that if site X is showing a banner ad from an advertiser (A), then advertiser A can write cookies onto the host machine, just because of the presence of an ad on site X. Additionally, if A is also advertising on site Y, then A can use the same cookie used on site X. This allows A to track you, personally, identifiable by an IP address, across all the Web Sites that it advertises on. This simulation from Privacy.net shows what kind of data a cookie can record across sessions on different domains. Here, the country of origin, the IP address, a timestamp and



An example of how cookies are used to track users

the web page viewed is displayed.

Toolbars and add-ons for your browser can track your activities on all the pages you visit, and even go as far as monitoring the cookies that are saved by all these pages on your machine. Google's toolbar does this, something hidden away in the agreement. Google also uses an infamous DoubleClick cookie to track you across all the pages that advertise using Google's ad-sense. Google claims to do this to "improve your online experience, including the display of more relevant ads". Fortunately, you are allowed to opt-out, and disable the cookie by downloading and installing a plug-in available at <http://www.google.com/ads/preferences/plugin/>. The opt-out

approach for disabling cookies, is the de facto standard of the web, as against an opt-in approach, which is more acceptable for privacy activists. This was the major cause for the privacy hullabaloo around Google's Buzz service, as users

an individual search history of all your searches. If you have a Google account, then all your searches are saved, which means that any potentially embarrassing searches you have made are available for those who want to view it. You can disable this tracking by going to the Search History tab on the main page of Google.

Companies are legally required to monitor and track activities. If you have an e-mail, instant messenger, or a "group" id, then your activities are being monitored in case they are necessary in legal proceedings. Typically, access logs are kept for 60 to 90 days, of every IP address that was used to sign in to an account. A service such as Hotmail is legally required to provide the e-mails

Start Time UTC	End Time UTC	Gamer Tag	Title Name	IP
10/29/2005 08:59 PM	10/29/2005 09:00 PM		Xbox Dashboard	
10/29/2005 09:01 PM	10/29/2005 10:39 PM		SW: Jedi Academy	
10/29/2005 10:39 PM	10/29/2005 10:40 PM		Xbox Dashboard	
10/29/2005 10:41 PM	10/29/2005 10:43 PM		Midnight Club 3: DUB	
10/29/2005 10:43 PM	10/29/2005 10:44 PM		Xbox Dashboard	
10/29/2005 10:45 PM	10/29/2005 10:56 PM		Midnight Club 3: DUB	
10/30/2005 01:34 AM	10/30/2005 05:02 AM		Midnight Club 3: DUB	
10/31/2005 03:36 AM	10/31/2005 04:08 AM		Midnight Club 3: DUB	
10/31/2005 12:33 PM	10/31/2005 12:36 PM		Midnight Club 3: DUB	
10/31/2005 09:28 PM	10/31/2005 10:10 PM		Midnight Club 3: DUB	

An example of an XBOX Live activity report

in the account if a proceeding of law requires it. Logs are kept for multi-player gaming on consoles as well. Shown above is an example of what an XBOX Live activity log looks like.

Social Networking

Social networking sites have a rich database of individual choices and preferences, but fortunately, the scope of current computational methods to distill this information for commercial purposes is very limited. The approaches that analytics services use to extract useful information from visitor activity on web sites can be used far more effectively on the wealth of information available if user interaction on social networking sites are tracked. Social networking sites have a unique problem

facebook

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had to opt-out of displaying their closest Gmail contacts on the new Buzz profile, instead of opting in.

Search engines are also uniquely in position to track and harvest search queries. Services like trends anonymously use such data, but Google maintains